Harmful Algae 14 (2012) 36–45

Contents lists available at SciVerse ScienceDirect

Harmful Algae

journal homepage: www.elsevier.com/locate/hal

Ecosystem disruptive algal blooms of the brown tide species, *Aureococcus anophagefferens* and *Aureoumbra lagunensis*

Christopher J. Gobler [a,*], William G. Sunda [b]

[a] School of Marine and Atmospheric Sciences, Stony Brook University, Stony Brook, NY 11790-5000, USA
[b] National Ocean Service, NOAA, 101 Pivers Island Road, Beaufort, NC 28516, USA

ARTICLE INFO

Article history:
Available online 25 October 2011

Keywords:
Aureococcus, *Aureoumbra*, Brown tide, Algal blooms, Nutrients, Grazing

ABSTRACT

Brown tides of the pelagophytes *Aureococcus anophagefferens* Hargraves et Sieburth and *Aureoumbra lagunensis* DeYoe et Stockwell have formed ecosystem disruptive algal blooms in shallow estuaries of the United States and South Africa for more than two decades. These algae achieve maximum growth rates at low light and nutrient levels and are able to utilize a wide variety of organic nutrient (N and P) substrates. They also experience low mortality losses via their ability to discourage grazing by zooplankton and filter feeding mollusks. Observations and conceptual and numerical models have indicated the importance of 'pre-blooms' of competing high nutrient-adapted phytoplankton that reduce available nutrients to low levels prior to the formation of many brown tide blooms, especially those of *Aureococcus*. These models suggest that brown tides are promoted by positive feedback mechanisms involving the ability of these algae to grow competitively at low nutrient and light levels, their low rates of grazing mortality, and the associated low grazing-mediated recycling of nutrients. The latter further reduces nutrient concentrations, which further favors competitive population growth of the brown tide species, thereby promoting blooms. Molecular and genomic studies of brown tides have identified the genes that encode many of the key proteins that facilitate bloom formation and persistence. Brown tides are among the most widely studied harmful algal blooms (HABs), which makes them ideal model systems for identifying the factors and complex mechanisms that promote HAB events. As such, several promising bloom mitigation strategies are now apparent.

© 2011 Elsevier B.V. All rights reserved.

1. Background

Brown tides are caused by the pelagophytes *Aureococcus anophagefferens* Hargraves et Sieburth and *Aureoumbra lagunensis* DeYoe et Stockwell (Fig. 1). *Aureococcus* has caused destructive brown tide blooms in northeast and mid-Atlantic U.S. estuaries for more than 25 years (Nuzzi and Waters, 2004), and more recently in South Africa (Probyn et al., 2001, 2010). *Aureoumbra* caused a massive continuous bloom in coastal bays in and around the Laguna Madre, Texas from 1990 to 1997 and intermittent blooms in subsequent years (Buskey et al., 2001; DeYoe et al., 2007). Neither species was known to science prior to the first documented bloom events nor affects human health, but both have since gained notoriety for their ability to disrupt and damage the coastal ecosystems in which they occur.

Although *Aureococcus* and *Aureoumbra* are genetically distinct organisms (DeYoe et al., 1995), they share many similarities. Both contain the pigment 19′-butanoyloxyfucoxanthin as their major carotenoid and are small (4–5 μm for *Aureoumbra* and 2–3 μm for *Aureococcus*), spherical, non-motile cells with simple life cycles (DeYoe et al., 1997). They were originally assigned to the class Chrysophyceae (Sieburth et al., 1988; Buskey and Stockwell, 1993), but later examination of their pigments, physiology, 18S rRNA sequences and morphology led to their formal placement in the class Pelagophyceae (DeYoe et al., 1997). Both contain the pelagophyte-specific sterol, (Z)-24-propylidenecholesterol, which further supports this classification (Giner et al., 2001). Also as we describe in this review, these two species share many physiological attributes that promote the development and persistence of brown tide blooms.

The first *Aureococcus* blooms occurred simultaneously in the summer of 1985 in several estuaries in the northeastern U.S., including Narragansett Bay, Rhode Island, Great South Bay, and the Peconic Estuary on Long Island, NY, and putatively in Barnegat Bay, NJ (Sieburth et al., 1988; Cosper et al., 1989; Olsen, 1989). Since then blooms have occurred almost annually in various in Long Island bays. Blooms have not occurred in the Peconic Estuary since 1995, although low cell densities are still present (up to 10^4 ml^{-1}; Nuzzi and Waters, 2004). By contrast, *Aureococcus* continues to

* Corresponding author.
E-mail addresses: cgobler@notes.cc.sunysb.edu (C.J. Gobler), bill.sunda@noaa.gov (W.G. Sunda).

1568-9883/$ – see front matter © 2011 Elsevier B.V. All rights reserved.
doi:10.1016/j.hal.2011.10.013

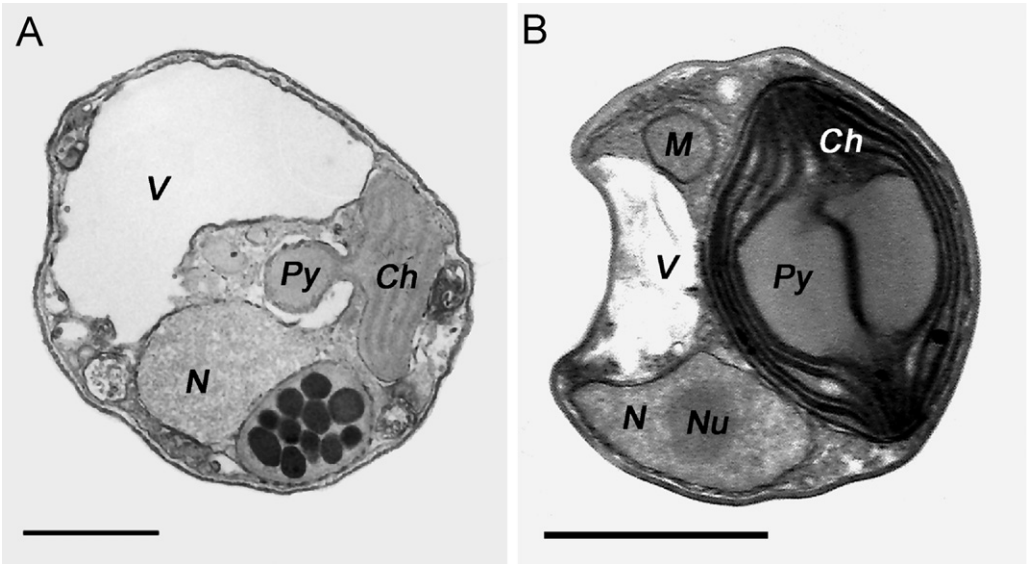

Fig. 1. Transmission electron micrographs of *Aureoumbra lagunensis* (A) and *Aureococcus anophagefferens* (B) showing the nucleus (N), nucleolus (Nu), chloroplast (Ch), pyrenoid (Py), mitochondrion (M), and vacuole (V). Scale bars = 1 μm. Images courtesy of O. Roger Anderson, Columbia University.

form dense blooms within the south shore lagoons of Long Island (Great South, Moriches, Shinnecock, and Quantuck Bays; Gobler et al., 2011). While *Aureococcus* has been observed in Narragansett Bay in subsequent years, it has not formed a significant bloom there since 1985 (Sieburth et al., 1988). In recent years, blooms have expanded south along the U.S. east coast into bays in New Jersey (Barnegat Bay; Gastrich et al., 2004), Delaware (Little Assawoman Bay; Popels et al., 2003), Maryland, and Virginia (Chincoteague Bay; Trice et al., 2004; Glibert et al., 2007; Mulholland et al., 2009). *Aureococcus* blooms have also recently occurred in Saldanha Bay, South Africa (Probyn et al., 2001, 2010). Low abundances of *Aureococcus* cells have been observed along the entire eastern seaboard of the U.S. from Maine to Florida, indicating the potential for these blooms to expand north and south of their current range (Anderson et al., 1993; Popels et al., 2003).

The initial *Aureoumbra* brown tide bloom began in shallow lagoons in and around the Laguna Madre and Baffin Bay, Texas in January 1990 and persisted for nearly eight years, making it the longest continuous harmful algal bloom (HAB) event ever recorded (Buskey et al., 1997, 2001). During this time, *Aureoumbra* comprised most of the algal biomass and generally maintained cell densities from 0.5 to 5×10^6 ml^{-1} (Buskey and Stockwell, 1993; Buskey et al., 1997). The bloom terminated in the fall of 1997, but re-emerged during the summer and fall of 1998 (Buskey et al., 2001). Since then brown tide blooms have occurred intermittently in the Laguna Madre system, but have been of comparatively short duration and limited spatial extent (DeYoe et al., 2007). Low concentrations of *Aureoumbra* cells have also been found in coastal bays across Florida, Texas and Mexico (Villareal et al., 2004).

During the initial *Aureococcus* blooms in eastern and southern Long Island bays (1985–1987), high cell concentrations ($>10^6$ ml^{-1}) substantially increased light attenuation, which caused a large-scale die-off of seagrass beds of *Zostera marina*, a critical habitat for scallops, larval fish, and other species (Cosper et al., 1987). The blooms caused mass mortality and recruitment failure in populations of bay scallops (*Argopecten irradians*), which resulted in the collapse of the multi-million dollar scallop industry in eastern Long Island (Bricelj and Kuenstner, 1989). The *Aureococcus* blooms also negatively impacted populations of clams (*Mercenaria mercenaria*) in Great South Bay, which was formerly the most valuable fishery in the state of New York. While overharvesting was identified as the cause of a sharp reduction in clam populations during the early 1980s, chronic *Aureococcus* blooms during the late 1980s and early 1990s contributed toward the 99% reduction in landings that occurred during this period (Kraeuter et al., 2008).

Subsequent research has demonstrated that *Aureococcus* adversely impacts the growth, survival, and reproduction of many algal grazers, including larval, juvenile and adult hard clams (*M. mercenaria*; Bricelj et al., 2001; Greenfield and Lonsdale, 2002; Padilla et al., 2006; Bricelj and MacQuarrie, 2007; Newell et al., 2009), larval and adult bay scallops (*A. irradians*; Bricelj and Kuenstner, 1989; Gallager et al., 1989), adult and juvenile blue mussels (*Mytilus edulis*; Bricelj and Kuenstner, 1989; Bricelj et al., 2001), and micro- and mesozooplankton (Lonsdale et al., 1996; Caron et al., 2004; Deonarine et al., 2006; Smith et al., 2008). Although a toxin has never been isolated from *Aureococcus*, there is evidence for toxic activity, most likely within the extracellular polysaccharide (EPS) sheath surrounding the cells (Sieburth et al., 1988). This putative toxin deters feeding in bivalves by stimulating contractions of the branchial muscles and interfering with lateral and ventral groove ciliary beating (Gainey and Shumway, 1991; Robbins et al., 2010). In addition, it has also been argued that *Aureococcus* may be a poor source of nutrition for zooplankton (Lonsdale et al., 1996; Caron et al., 2004) and some bivalve larvae (Padilla et al., 2006; Bricelj and MacQuarrie, 2007). The loss of toxicity of one strain of *Aureococcus* (CCMP 1784) since its isolation permitted Bricelj et al. (2004) to compare toxic and non-toxic clones of *Aureococcus* and unambiguously demonstrate that the detrimental effects of *Aureococcus* on bivalve growth are mainly attributable to cell toxicity, rather than high cell density, nutritional deficiency, or poor filtration retention of the small cells.

The ecological impacts of *Aureoumbra* appear to have been similar to those of *Aureococcus*, although the economic impacts have been lower because of a less-developed shellfish industry in and around Laguna Madre. As with *Aureococcus*, the *Aureoumbra* bloom caused a substantial increase in light attenuation, and the resulting shading of the bottom decreased the abundance of the once extensive sea grass beds (Onuf, 1996). There was also a decrease in the biomass and diversity of benthic invertebrates in the Laguna Madre (Ward et al., 2000). The dominant clam, *Mulinia lateralis*, virtually disappeared after the onset of the brown tide

bloom (Montagna et al., 1993), and the dominate polychaete, *Streblospio benedicti*, an important grazer of phytoplankton, declined in abundance by two orders of magnitude (Buskey et al., 1997). However, population decreases in both species may have begun during the period of high salinity and freezing temperatures prior to the bloom (Buskey et al., 1997). The brown tide bloom also was associated with substantial decreases in the grazing activity, growth, and egg release rates of mesozooplankton (e.g., *Acartia tonsa*), and decreases in the abundance and grazing rates of microzooplankton (Buskey and Stockwell, 1993).

Like *Aureococcus*, an EPS polymer sheath covers the cell surface of *Aureoumbra*. In feeding experiments with *Aureoumbra*, the presence of high levels of this exopolymer decreased rates of grazing and/or growth in three protozoan species and altered protozoan motility (Liu and Buskey, 2000). Liu and Buskey speculated that these effects on grazing and motility may have been caused by the adherence of the exopolymer to cilia on the surface of the protozoans. These cilia are essential for protozoan feeding and motility. Alternatively, the adverse effects may be caused by an unidentified toxin within the polysaccharide sheath, as has been suggested for *Aureococcus*. In addition, the polymer sheath protects *Aureoumbra* from digestion in the guts of zooplankton (e.g., copepods), and allows viable cells to pass through the gut unharmed (Bersano et al., 2002). This effect not only lessens grazing mortality losses but also stimulates absorption of nutrients released during the digestion of other food items (Porter, 1976), which could promote brown tide growth.

2. Nutritional and physical factors promoting brown tides

Although harmful algal blooms in coastal waters have been commonly attributed to nutrient loading, the role of nutrients in the occurrence of brown tides appears to be more complex than a simple nutrient stimulation of brown tide growth (Sunda et al., 2006). An examination of spatial and temporal patterns of concentrations of *Aureococcus* cells and inorganic nutrients indicated that blooms occurred when inorganic nutrient levels were low (Cosper et al., 1989; LaRoche et al., 1997; Fig. 2). Moreover, nitrate additions during laboratory, mesocosm, and bottle experiments in the field have consistently yielded reduced *Aureococcus* growth rates relative to those of competing algae (Keller and Rice, 1989; Gobler and Sañudo-Wilhelmy, 2001a; Gobler et al., 2002, 2004a,b; Taylor et al., 2006). The nitrogen-uptake characteristics of *Aureococcus* (low K_s and V_{max} for ammonium) suggest that this species is well adapted to low nutrient environments (Lomas et al., 1996). The ability of *Aureococcus* to attain high biomass levels when inorganic nutrient concentrations are low may be partly linked to its ability to utilize organic forms of C, N, and P. Cultures (both axenic and non-axenic) and field populations of *Aureococcus* have been shown to obtain nitrogen from a variety of organic compounds, including urea, amino acids, proteins, chitobiose, and acetamide (Berg et al., 2002; Mulholland et al., 2002, 2009), and to assimilate organic carbon from glucose and amino acids (Dzurica et al., 1989; Mulholland et al., 2002, 2009). Experimental additions of dissolved organic matter (DOM; glucose and amino acids) have enhanced the growth and relative abundance of *Aureococcus* during field experiments (Gobler and Sañudo-Wilhelmy, 2001a). Dissolved organic nitrogen (DON) and carbon (DOC) are often elevated during bloom initiation (LaRoche et al., 1997; Gobler et al., 2002, 2004b) and decreases in DOC and DON concentrations have been observed with bloom development, a pattern consistent with utilization of these substances by *Aureococcus* (LaRoche et al., 1997; Gobler et al., 2004b; Gobler et al., 2011) (Fig. 2). In Chincoteague Bay (MD), the concentrations of DON doubled from 1996 to 2004 during which time *Aureococcus* blooms began and intensified with increasing

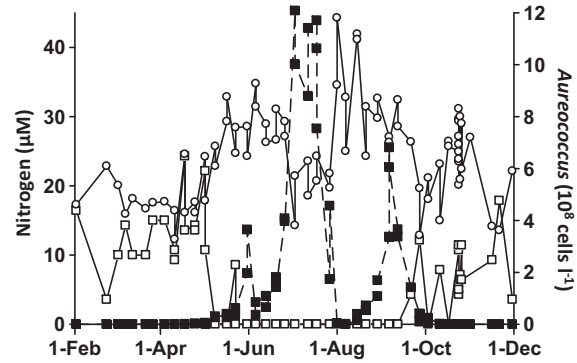

Fig. 2. Concentrations of nitrate plus nitrite (×10 □), dissolved organic nitrogen (○), and *A. anophagefferens* cells (■) in surface waters of Flanders Bay, Long Island, NY, USA, 1995. Figure from Sunda et al., 2006, based on data supplied by the Suffolk County Department of Health Services.

DON levels (Glibert et al., 2007). While DON provides an important N source to blooms, heterotrophic utilization of carbon may partly circumvent the need for photosynthetic carbon fixation, giving *Aureococcus* a competitive advantage over strictly autotrophic algae under severe shading during blooms.

Blooms of *Aureococcus* often occur after 'pre-blooms' of other algae, which both draw down inorganic nutrients to low levels and cause gradual increases DON and DOC pools that are less utilizable by other phytoplankton (Gobler and Sañudo-Wilhelmy, 2001b; Gobler et al., 2011) (Fig. 2). Microbial degradation processes during and following these pre-blooms can result in enhanced DOM levels, which can serve as a source of DON and DOC to subsequent *Aureococcus* blooms (LaRoche et al., 1997; Gobler and Sañudo-Wilhelmy, 2001b; Glibert et al., 2007). Since these blooms usually occur in shallow bays, benthic fluxes are important sources of inorganic and organic nutrients (Lomas et al., 2004). In experiments, the enrichment of bloom water with benthic pore water significantly increased the absolute and relative abundance of *Aureococcus* (Taylor et al., 2006). Whether sediments serve as a net source or sink for nutrients often depends on the degree to which benthic primary producers (microalgae, sea grass, or macroalgae) utilize nutrients regenerated in sediments (MacIntyre et al., 2004). Brown-tide shading of the bottom causes a net loss of these benthic phototrophs, and this loss provides a source of nutrients and DOM via plant matter decay and recycling processes. It also increases net benthic flux of nutrients to the water column due to decreased utilization by benthic phototrophs. Both processes should promote the development of brown tide blooms (MacIntyre et al., 2004).

Considerably less is known about the role of nutrients in the development and persistence of *Aureoumbra* blooms. The initial brown tide bloom in the Laguna Madre and Baffin Bay, Texas developed after an extremely dry period, and a rare freeze in December 1989 (Buskey et al., 1997). The freeze caused extensive mortalities of fish and other organisms, and a subsequent large pulse of regenerated ammonium (up to 25 μM; Buskey et al., 1996). The ammonium pulse stimulated the growth of *Aureoumbra* to bloom levels (~10^6 cells l^{-1}) in the upper reaches of Baffin Bay during January 1990, but the bloom did not spread to lower Baffin Bay or the adjacent Laguna Madre until June of that year (Buskey et al., 1997). Thereafter, ammonium concentrations plummeted to very low levels (<1 μM) in the Laguna Madre system and remained low during the following year, likely due to the large demand created by the high *Aureoumbra* biomass (Buskey et al., 1996).

There is evidence that *Aureoumbra* is well adapted to low concentrations of available nutrients. Chemostat studies indicate that it can grow at low cellular N:C and P:C ratios (Liu et al., 2001;

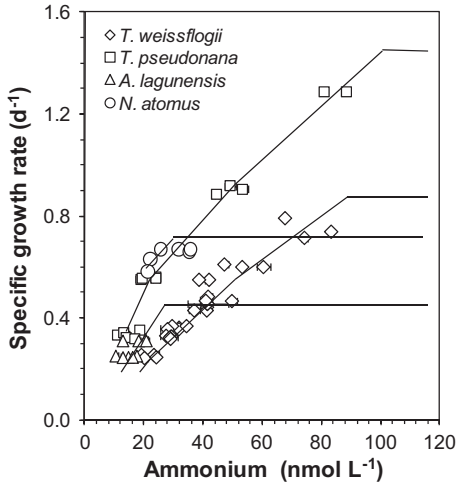

Fig. 3. Specific growth rate vs dissolved [NH_4^+] for EDAB species *Aureoumbra lagunensis* (4–5 μm diameter), *Nannochloris atomus* (3 μm), and the diatoms *Thalassiosira weissflogii* (10–11 μm), and *T. pseudonana* (4–5 μm). Data were measured in N-limited continuous cultures at 20 °C under a light/dark cycle of 14 h/10 h. Maximum growth rates measured in nutrient-sufficient batch cultures are indicated by horizontal lines. Note that *Aureoumbra* and *Nannochloris* reach maximum growth rates at low ammonium concentrations, while that of the diatoms require much higher concentrations to reach maximum rates. Consequently the EDAB species can effectively compete for growth only at low ammonium concentrations.

Sunda and Hardison, 2010), and that its growth rate becomes ammonium-limited only at concentrations below ~0.03 μM, much lower than threshold values (0.08–0.10 μM) for ammonium limitation of diatoms and other high-nutrient adapted species (Fig. 3) (Sunda and Hardison, 2007, 2010). Moreover, its small cell size, like that of *Aureococcus*, results in a high surface to volume ratio and a thin surface diffusive boundary layer, which further favors growth of this species at low nutrient concentrations (Raven and Kubler, 2002; Sunda and Hardison, 2007, 2010). *Aureoumbra* utilizes ammonia and nitrite as inorganic nitrogen sources, but is unable to grow on nitrate, usually the most available nitrogen source under high-nutrient conditions (DeYoe and Suttle, 1994). Measurements of particulate C:N molar ratios (28 ± 8) and cellular chlorophyll *a* (0.019 ± 0.002 pg $cell^{-1}$) during the early stages of the Texas brown tide bloom in August 1990 yielded values indicative of nitrogen limited cells (DeYoe and Suttle, 1994). In a mesocosm study during a subsequent bloom in 1999, the addition of 40 μM ammonia increased the size and chl fluorescence of *Aureoumbra* cells, again consistent with nitrogen limitation of the ambient population, but there was no measurable increase in the net reproduction rate of brown tide cells (Buskey et al., 2003; DeYoe et al., 2007). Like *Aureococcus* and other low-nutrient adapted species, *Aureoumbra* is able to grow on organic nitrogen substrates, including urea and glutamate, albeit at maximum specific rates that are 40–50% lower than growth rates on ammonium (Muhlstein and Villareal, 2007). It is also able to utilize organic phosphate esters such as β-glycerophosphate (Muhlstein and Villareal, 2007), but its ability to assimilate and grow on organic carbon substrates has not been examined.

Physical factors (salinity, temperature, light, and water residence times) can also influence brown tide blooms. Both *Aureococcus* and *Aureoumbra* blooms are associated with high chlorophyll *a* (chl *a*) concentrations (30–60 μg l^{-1}) and associated severe light attenuation. They are both able to maintain near-maximum growth rates (at 20 °C) at a light intensity of 50 μmol photons m^{-2} s^{-1} (~2% of noon solar irradiance; MacIntyre et al., 2004; Sunda and Hardison, unpublished data). *Aureococcus* blooms have been shown to display significant N uptake in the dark (Probyn et al., 2010) and *Aureococcus* cultures can survive two weeks of complete darkness and can then rapidly resume growth (Popels et al., 2007). In addition, the shallow (<50 cm) photic zone observed during *Aureococcus* blooms combined with wind mixing creates large fluctuations in the light field experienced by cells, a condition that significantly increases the short-term carbon fixation and photosynthetic efficiency in *A. anophagefferens* compared to constant irradiance (Milligan and Cosper, 1997). Curiously, the accessory pigments in *Aureococcus* make it best adapted for growth at low levels of blue light, as found in the deep chlorophyll maximum in the open ocean where pelagophytes are commonly found (Goericke, 1998). It is less-well-adapted for growth at low intensities of the longer wavelength visible light that occur during brown tide blooms (Yentsch et al., 1989). Based on these observations, Yentsch et al. suggested that *Aureococcus* might be an expatriate open-ocean species. The ability of the two brown-tide species to grow maximally at low light intensities may contribute to their ability to compete well with non-HAB species during blooms (Milligan and Cosper, 1997; MacIntyre et al., 2004).

Aureococcus grows optimally at salinities ≥24 and intense blooms of this species are more likely to occur during dry years with higher salinities (Cosper et al., 1989; LaRoche et al., 1997). This trend, however, is more likely related to the reduced inorganic nutrient loading from lower groundwater inputs and to increased bay water residence times (see below), rather than salinity *per se* since the salinities of most bays that experience brown tide blooms rarely fall below 24 (Cosper et al., 1989; LaRoche et al., 1997; Gobler and Sañudo-Wilhelmy, 2001b). Experimental incubations of bloom water mixed with small amounts of nitrate-rich groundwater have been shown to reduce the relative abundance of *Aureococcus* within the phytoplankton community (Taylor et al., 2006), supporting the idea that increased groundwater inputs discourage brown tide blooms by encouraging the growth of competing algae, at least on an immediate short-term basis.

Aureoumbra can grow over a very wide range of salinities (10–90) and maintains near optimal growth at salinities of 20–70 at 25 °C (Buskey et al., 1998). This ability to acclimate to a broad salinity range and to grow well at very high salinities may have contributed to the initiation of the brown tide bloom in Laguna Madre (Buskey et al., 1998). The Laguna Madre estuary is a shallow (1.2 m average depth) hypersaline lagoon within a semi-arid region of south Texas. Salinities can vary considerably, depending on rainfall, but the initial *Aureoumbra* bloom occurred after a prolonged dry period. It was initiated in the hypersaline upper reaches of Baffin Bay in January 1990, where maximum salinities reached 70 psu. This salinity was within the range for near-optimal growth of *Aureoumbra*, but well above the optimal range for most algal grazers and competing algal species (Buskey et al., 1998). During the period just before the bloom there was a significant negative correlation between microzooplankton abundance and salinity, with very low abundances (<20 ml^{-1}) at the highest salinities (66–72). The low growth rates of competing algae and low grazing pressure on *Aureoumbra* associated with the high salinities appeared to be important contributing factors in bloom initiation (Buskey et al., 1998).

The wide salinity tolerance of *Aureoumbra* also likely contributed to the unprecedented persistence of the bloom. The bloom terminated in the fall of 1997, after intense rains decreased average salinities in Baffin Bay from ~50 to 15, below the range for optimal *Aureoumbra* growth (Buskey et al., 2001). Thus, the demise of the bloom may have been partly caused by a decrease in brown tide growth and a concomitant increase in the growth rate of competing phytoplankton associated with the lower salinities. In addition, an influx of nutrients accompanying the large freshwater input stimulated a bloom of high-nutrient adapted algae (the diatom *Rhizosolenia* sp.), which was then succeeded by the re-emergence of a brown tide bloom (Buskey et al., 2001).

Aureococcus blooms under a wide range of temperatures (0–25 °C) and shows optimal growth at ~20 °C (Cosper et al., 1989; Gobler et al., 2002). Blooms typically initiate during May or June as temperatures approach 15–20 °C, and decline when temperatures exceed 25 °C during July or August (Nuzzi and Waters, 2004). *Aureococcus* blooms can also re-emerge and even initiate during the fall when temperatures fall below 20 °C, and can persist during winter when temperatures reach freezing (Gobler et al., 2002; Nuzzi and Waters, 2004).

Temperature may also influence *Aureoumbra* bloom dynamics. Temperatures in the Laguna Madre usually range from 10 to 15 °C in the winter to ~30 °C in summer, within a favorable range for *Aureoumbra* (Buskey et al., 1996, 1998). In culture studies, the specific growth rate of *Aureoumbra* increased with temperature within the seasonal range, with a rate of 0.25 d^{-1} at 15 °C, 0.62 d^{-1} at 25 °C, and 0.87 d^{-1} at 30 °C (Buskey et al., 1998; Sunda and Hardison, unpublished data). Thus, unlike *Aureococcus*, the highest summer temperatures in the Laguna Madre system appear to favor rapid growth of *Aureoumbra*. The ability to grow over a wide range of temperatures and salinities likely contributed to the unprecedented seven year persistence of the initial Texas brown tide bloom.

While bays that host frequent *Aureococcus* blooms are known to have high salinities and are relatively shallow, an additional attribute of these bays is their long residence times (up to two months), which are associated with low freshwater inputs and low rates of mixing with coastal seawater (Hardy, 1976; Wilson, 1995). Similarly, estimates of residence times for the upper Laguna Madre and Baffin Bay, Texas, which host *Aureoumbra* blooms, can range from 300 days to several years (Buskey et al., 1998, 2001). While long residence times clearly permit the accumulation of high algal biomass, they would not necessarily give a competitive advantage to one algal species over another. However, the high biomass and low inputs of external nutrients in these systems create large algal nutrient demand:supply ratios, and hence low concentrations of available inorganic nutrients. Moreover, intense nutrient recycling within these shallow, long-residence time systems promotes the accumulation of DOM (Lomas et al., 2004). The resulting low concentrations of available inorganic nutrients and low ratios of inorganic to organic pools of N and P likely favor the net population growth of brown tide species.

3. Sources of cell mortality

Because of their small size, *Aureococcus* and *Aureoumbra* should be efficiently grazed by micro- and nanozooplankton, whose rapid growth rates should prevent blooms from developing (Sherr and Sherr, 2002). However, grazing rates on these species are typically low due to unpalatability, toxicity, or physical interference with grazing. These low rates of grazing mortality have been proposed to be important factors in the initiation and persistence of brown tide blooms (Buskey et al., 1997; Gobler et al., 2002; Sunda et al., 2006; Smayda, 2008). While zooplankton are able to actively graze during *Aureococcus* blooms, specific grazing rates on *Aureococcus* can be considerably lower (e.g. 70% lower; Gobler et al., 2002) than grazing rates on competing algae, and lower than concurrent cellular growth rates for *Aureococcus* (Gobler et al., 2002, 2004a; Caron et al., 2004). Reduced grazing may be partly facilitated by the EPS layer of *Aureococcus*. Another mechanism that may decrease grazing on *Aureococcus* is a microbial trophic cascade in which enhanced grazing by larger zooplankton reduces microzooplankton populations, which in turn reduces grazing on brown tide cells (Sieracki et al., 2004). However, results of a mesocosm study suggest that this mechanism does not promote Texas brown tide blooms (Buskey et al., 2003).

Aureococcus blooms have waned in their intensity and frequency in some U.S. estuaries such as the Peconic Estuary, NY. This system has not experienced changes in its chemical or physical characteristics relative to periods when intense blooms first occurred (Nuzzi and Waters, 2004). Grazing communities can evolve resistance to harmful algae over time via natural selection processes (Hairston et al., 2002). The selective adverse impact of blooms on more sensitive grazer species and more sensitive phenotypes within populations (Caron et al., 2004) should eventually lead to the establishment of grazer communities that are better adapted to co-exist with and consume *Aureococcus*. Recent studies have shown that some NY estuaries, which formerly hosted zooplankton communities that consumed *Aureococcus* at low rates during massive blooms, have shifted to communities that actively graze *Aureococcus* at rates similar to those for other algal species (Deonarine et al., 2006). Similarly, although *Aureococcus* inhibits grazing by mollusks (Bricelj and Kuenstner, 1989; Bricelj et al., 2001; Gallager et al., 1989), the filter feeding slipper limpet, *Crepidula fornicata*, was able to avidly graze multiple toxic strains and natural bloom samples of *Aureococcus* while other mollusks (e.g. *Mercenaria mercenaria*) could not (Harke et al., 2011). This limpet is now present at high abundance in estuaries that no longer experience brown tide blooms (e.g., the Peconic Estuary; Harke et al., 2011). Thus, the recent abatement of brown tides in some Long Island bays may, at least partly, reflect a shift in the grazing communities toward populations more capable of feeding on *Aureococcus*.

The occurrence of brown tides in Long Island bays may be partly related to the dramatic decline in shellfish populations, initially from over-harvesting and later from mortality caused by persistent brown tide blooms (Kraeuter et al., 2008). In support of this hypothesis, mesocosms containing high but environmentally realistic concentrations of hard clams (*M. mercenaria*) maintained non-bloom densities of *Aureococcus* (~10^3 ml^{-1}), while identical mesocosms with few or no clams developed dense brown tide blooms (>10^5 ml^{-1}) (Cerrato et al., 2004). *Aureococcus* inhibits *M. mercenaria* filtration rates at densities above 4×10^4 cells ml^{-1} (Bricelj et al., 2001). Thus, there may be a threshold effect whereby clams consume *Aureococcus* at low cell densities, thereby retarding bloom development, but their grazing is inhibited at high cell concentrations, leading to the formation of dense brown tides (Bricelj et al., 2001).

Reduced grazing pressure appears to have been an important factor in both the initiation and persistence of the eight-year Texas brown tide bloom (Buskey and Stockwell, 1993; Buskey et al., 1997). As noted earlier, the bloom was initiated in the hypersaline upper reaches of Baffin Bay and was preceded by a period of unusually high salinities (up to 70 psu) that greatly reduced the abundance of microzooplankton and important benthic grazers such as the bivalve *Mulinia lateralis* (Buskey et al., 1997). The spread of the bloom to lower Baffin Bay and the Laguna Madre in June 1990 was accompanied by large declines in both the abundance of mesozooplankton (e.g., *Acartia tonsa*) and in grazing rates of microzooplankton (Buskey and Stockwell, 1993). Gut fullness, growth, and egg laying in adult *Acartia tonsa* were severely reduced during the bloom, suggesting that the zooplankton were either unable to feed upon, or actively avoided *Aureoumbra* cells. Subsequent laboratory experiments verified these findings for adult *Acartia* and showed that the presence of *Aureoumbra* increased the mortality of *Acartia* nauplii relative to unfed (starved) nauplii (Buskey and Hyatt, 1995). Increasing concentrations of *Aureoumbra* also caused a dose-dependent increase in mortality in two microzooplankton species (the ciliate *Strombidinopsis* sp. and the rotifer *Brachionus plicatilus*). A third species was unable to grow on *Aureoumbra* and three others had reduced growth rates compared to cultures fed on equal concentrations of control algae (Buskey and Hyatt, 1995). The increased mortality observed in some zooplankton species suggests the presence of a

toxin, but no toxin has yet been identified (Liu and Buskey, 2000). The putative toxicity of *Aureoumbra* or an inability of zooplankton to feed and grow on this algal food source likely contributed substantially to the development and persistence of the long-lived Texas brown tide bloom (Buskey et al., 2001).

Viruses may also influence the dynamics of *Aureococcus* blooms. Electron micrographs of the first-observed brown tide event in Rhode Island and subsequent blooms in New Jersey and New York revealed the presence of intracellular, icosahedral virus–like particles in *Aureococcus* cells (Sieburth et al., 1988; Gastrich et al., 2002), suggesting that brown tide populations may experience viral infection and lysis. Viral densities during *Aureococcus* blooms are generally elevated compared to most estuarine environments (Gobler et al., 2004a) and field studies have observed a high percentage (~40%) of virally infected *Aureococcus* cells at the end of blooms, suggesting that viruses may be an important source of mortality during bloom termination (Gastrich et al., 2004). *Aureococcus*-specific viruses capable of completely lysing brown tide cultures have been isolated from bloom waters in Long Island estuaries (Milligan and Cosper, 1994; Gobler et al., 2007; Rowe et al., 2008). Axenic cultures of *Aureococcus* infected with field-isolated viruses have been shown to host large (140 nm) icosahedral viruses with morphologies consistent with *Phycodnaviridae* although contaminating bacterial phages were also found in some viral concentrates isolated from the field (Rowe et al., 2008). Collectively, these findings suggest that early reports and images of phage-like viruses infecting *Aureococcus* (Milligan and Cosper, 1994) were actually bacterial viruses infecting bacteria in the non-axenic cultures. Isolated *Aureococcus* viruses infect only a portion of brown tide strains examined (9 of 19; Gobler et al., 2007), suggesting that viruses may impact both bloom diversity and biomass. Cultured viruses isolated from NY coastal waters significantly reduced bloom densities of *Aureococcus* during short term bottle incubation experiments, suggesting that field populations are also susceptible to viral infection (Gobler et al., 2007). To date, there is no evidence for viruses that infect *Aureoumbra* despite efforts to identify virally infected cells by TEM during blooms (C.J. Gobler, pers. obs.).

Another source of mortality to brown tides is algicidal bacteria. Frazier et al. (2007) cultured six clonal isolates of bacteria from Great South Bay, NY, USA, that were capable of lysing cultured *Aureococcus* cells. The extent to which these bacteria are active during *Aureococcus* blooms and the existence of algicidal bacteria capable of lysing or restricting the growth of *Aureoumbra* have not been investigated.

Allelopathic interactions with micro- and macroalgae may inhibit the formation of brown tide blooms. For example, the toxic dinoflagellates *Cochlodinium polykrikoides* and *Alexandrium fundyense* both excrete allelochemicals that inhibit the growth of *Aureococcus* (Tang and Gobler, 2010; Hattenrath-Lehmann and Gobler, 2011). Interestingly, while all three species form HABs in NY estuaries, brown tides are temporally and spatially separated from blooms of these dinoflagellates. Similarly, the green macroalgae, *Ulva lactuca*, inhibited the growth of *Aureococcus* in field and lab experiments and *Aureococcus* blooms do not occur in shallow tidal creeks dominated by *Ulva* (Tang and Gobler, 2011).

4. Positive feedback interactions and brown tide initiation and persistence

Factors regulating harmful algal blooms, such as bottom up control by nutrients and top down control by grazers, have been typically considered separately with little regard for how these factors might interact. Recently, Sunda et al. (2006) proposed that monospecific blooms of brown tide (*Aureococcus* and *Aureoumbra*) and some other algal species that disrupt or degrade ecosystems, so-called ecosystem disruptive algal blooms (EDABs), may result from complex feedback interactions among nutrient regulation of algal growth, population losses to grazing, and grazer-mediated recycling of nutrients (Fig. 4A and B). In this model they noted that for a brown tide bloom to occur, the net population growth rate of the brown tide species must be positive, and must exceed that for competing algae (e.g., diatoms). Sunda et al. hypothesized that *Aureoumbra*, *Aureococcus* and other EDAB algae are low-nutrient adapted (*K*-selected) species that compete well with other phytoplankton only at low concentrations of available nutrients. This hypothesis was based on theoretical arguments, but has since been confirmed experimentally for *Aureoumbra* and other EDAB species (Sunda and Hardison, 2007, 2010; Fig. 3).

Sunda et al. further proposed that EDAB events usually require a pre-bloom of high-nutrient adapted species, such as diatoms, which draw down nutrients to low levels, and stimulate the population growth of grazer species. The pre-bloom also increases inputs of organic nitrogen and phosphorus compounds via grazer-mediated recycling. Sunda et al. argued that the combination of the ability to grow competitively at low nutrient availability and low

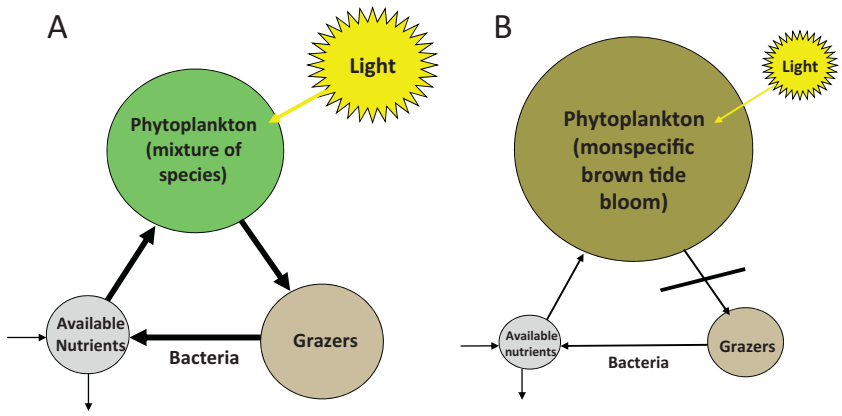

Fig. 4. Schematic diagrams for ecosystem interactions among limiting nutrients (e.g., N or P), the phytoplankton community, algal grazers (e.g., zooplankton), and bacteria. (A) Diagram for a healthy grazed, recycled system in which rates of nutrient uptake by phytoplankton, nutrient transfer to grazers, and subsequent nutrient recycling via grazer excretion and bacterial remineralization of organic matter roughly balance one another. In this recycled system (typical of shallow lagoons), rates of external nutrient inputs and loss (e.g. from advection) are small relative to internal cycling as indicated by the small size of the arrows entering and leaving the nutrient "box". (B) Diagram for a system that has been hijacked by a toxic or unpalatable EDAB species (e.g. a brown tide species) that severely disrupts algal grazing, and thus, diminishes nutrient supply via recycling. The decrease in recycled nutrient inputs decreases available nutrient concentrations, which favors the growth of the EDAB species, and also increases its toxicity or unpalatability. These effects result in a positive feedback that allows the EDAB species to dominate the algal community. Figure modified from Sunda et al. (2006).

grazing mortality rates would allow brown tide and other EDAB species to proliferate relative to populations of competing phytoplankton. As the competing phytoplankton are avidly grazed, the N and P from these algae would be recycled via zooplankton excretion and bacterial degradation processes to supply nutrients to fuel the growth of the emerging brown tide bloom (Fig. 4). While this occurs, the nutrient concentrations can be maintained at low levels via active algal uptake, which removes nutrients as fast as they are released from recycling. However, as the abundance of the brown tide species increases and that of the competing species declines, there will be a decrease in grazing rates, which decreases grazer-mediated recycling of nutrients. This decreased recycling decreases available nutrient concentrations, which further favors the population growth of the brown tide species relative to that of competing species. Sunda et al. hypothesized that these interactions constituted a positive feedback that promotes the formation of monospecific EDAB events, with co-occurring low concentrations of nutrients, algal grazers, and competing phytoplankton (Fig. 4B). Such exclusion of competing phytoplankton and grazers would also promote bloom persistence. These predictions have been confirmed in a recent brown tide bloom simulation model (Sunda and Shertzer, in press). In the model the biomass of pre-blooms and succeeding brown tide blooms increased proportionately with increasing concentrations of limiting nutrients entering into a coastal system, and thus, increased external nutrient inputs should increase the severity of brown tide blooms (Sunda and Shertzer, in press).

Sunda et al. (2006) noted that this positive feedback mechanism would be enhanced by increased toxin concentrations and other anti-grazer defenses, observed in many EDAB species under nutrient limitation. *Aureococcus* appears to fit this pattern as nutrient-limited, stationary phase cultures of this species cause greater inhibition of grazing rates in juvenile mussels, and growth rates in larval clams, than equal amounts of nutrient sufficient, exponentially growing cells (Bricelj et al., 2001; Padilla et al., 2006). This effect may be due to increased toxicity and/or increased production of extracellular polymer substances. As the nutrient concentrations are drawn down, first by the pre-bloom and then by growing demand from the emerging brown tide bloom, the grazing rates on the brown tide cells would progressively decrease, further favoring brown tide population growth and further decreasing nutrient recycling. The enhanced toxicity and grazer deterrence brought on by nutrient limitation provides a second positive feedback that promotes the formation and persistence of EDABs (Mitra and Flynn, 2006; Sunda et al., 2006).

Another positive feedback may be caused by benthic-pelagic coupling mechanisms (MacIntyre et al., 2004; Sunda et al., 2006). As noted above, blooms of *Aureococcus* and *Aureoumbra* cause high light attenuation, and thereby decrease the productivity and biomass of benthic algae and sea grass communities due to shading of the bottom (Cosper et al., 1987; Onuf, 1996). These benthic phototrophs compete with the brown tide species for nutrients, particularly those regenerated in the sediments, an important source of recycled nutrients in the shallow lagoons that host these blooms. The resultant decrease in the productivity and biomass of benthic phototrophs increases benthic nutrient inputs to the overlying water, further fueling bloom development and causing even more severe shading of the bottom and more light limitation of benthic phototrophs (MacIntyre et al., 2004). Benthic shading may also be exacerbated when the brown tide causes decreased filter feeding by shellfish (Montagna et al., 1993; Bricelj and Kuenstner, 1989; Bricelj et al., 2001), whose filtration clarifies the water column and thereby promotes benthic productivity. These benthic-pelagic coupling mechanisms constitute a third positive feedback that promotes the development and persistence of brown tide blooms, especially during later stages of blooms when light attenuation levels are high (Sunda et al., 2006). The three positive feedbacks should act in concert. However, the first two dealing with decreased grazing on brown tides and decreased grazer-linked nutrient recycling may tend to predominate during the earlier stages of the bloom, while the third feedback dealing with shading of the bottom and enhanced benthic inputs of nutrients should become increasingly important during the later bloom stages, and may be important in sustaining brown tide blooms.

5. Molecular studies of brown tides

With the introduction of high throughput genetic sequencing, understanding phytoplankton through a molecular and genomic lens has become increasingly common. A series of recent studies has moved *Aureococcus anophagefferens* to the forefront of genomic research for HABs, while some progress has been made with *Aureoumbra lagunesis* as well. These studies have enhanced our understanding of the physiological capabilities of brown tides and, to a lesser extent, their genetic diversity.

The genetic diversity of *Aureococcus* was initially investigated by Bailey and Andersen (1999) who sequenced the small subunit rRNA gene, the small subunit RUBISCO gene, and the non-coding spacer regions separating the RUBISCO subunit genes in 14 clones of *Aureococcus* isolated over a 12 year period (1986–1998) from three estuaries (the Peconic Bays and Great South Bay, NY and Barnegat Bay, NJ). This analysis revealed identical sequences for all genes among all clones of *Aureococcus* and led the authors to conclude that brown tides are monoclonal events (Bailey and Andersen, 1999). However, diversity inherently exists in all wild microbial populations, and the sequencing of additional genes or analyses of single-nucleotide polymorphisms or microsatellite markers are needed to better assess the true level of genetic diversity in *Aureococcus*. No study to date has characterized the genetic diversity of *Aureoumbra*.

Recently, the chloroplast genomes of both brown tide species were sequenced and compared to better understand the photosynthetic capabilities and low light adaptation in these algae (Ong et al., 2010). While this study confirmed the overall genetic similarities of these two species, it also demonstrated that the chloroplast genomes of both brown tide species are smaller than those of six other species of heterokonts whose chloroplasts have been sequenced (Ong et al., 2010). Additionally, three light-independent chlorophyll synthesis genes were found in the chloroplast of *Aureoumbra*, but not *Aureococcus*, suggesting a greater capacity for extended low light adaptation in *Aureoumbra*.

Aureococcus is the first eukaryotic HAB to have its entire genome sequenced and published (Gobler et al., 2011). The study generated metaproteomes of plankton assemblages present during brown tide blooms and identified six algal species that compete directly with *A. anophagefferens* in its natural environment. Gene sets involved in nutrient acquisition, light harvesting, and potential grazing defense were then compared among the seven species. Consistent with some of the physiological characteristics discussed in this review, *Aureococcus* contained more genes involved in the transport and metabolism of organic carbon and nitrogen compounds and the harvesting of light than the competing phytoplankton. A second finding consistent with prior studies was the enrichment in putative genes involved in the production of secondary metabolites that may encode for the synthesis and metabolism of toxins or other compounds that deter grazing (Fig. 5). Finally, the *Aureococcus* genome has more genes encoding for selenium-, copper-, and nickel-containing proteins than competing phytoplankton, a new finding for this alga. Recent studies have linked anthropogenic nutrient loading to HABs (Heisler et al., 2008). Given the large number of genes involved in adaptation to low light, assimilation of organic compounds, and

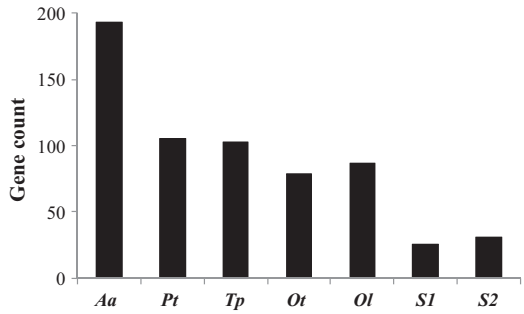

Fig. 5. Genes associated with secondary metabolite biosynthesis, transport, and catabolism in *A. anophagefferens* and six comparative phytoplankton as identified via KOG analysis (Eukaryotic Orthologous Groups) for identifying ortholog and paralog proteins. Aa, Pt, Tp, Ot, Ol, S1, and S2 are *Aureococcus anophagefferens*, *Phaeodactylum tricornutum*, *Thalassiosira pseudonana*, *Ostreococcus tauri*, *Ostreococcus lucimarinus*, *Synechococcus* clone CC9311 and *Synechococcus* clone CC9902, respectively.

synthesis of metalloproteins, anthropogenic activities that increase turbidity and concentrations of dissolved organic matter and trace metals in coastal ecosystems were hypothesized to promote brown tides and perhaps other HABs.

While genomic studies have provided insight into the genetic potential of brown tides, studies exploring gene expression can indicate the genes that are regulated in response to specific environmental conditions and that may help facilitate bloom formation. Berg et al. (2008) identified a series of trans-membrane transporters in *Aureococcus* that were up-regulated during N-limited growth on different N-containing nutrients, including urea, nitrate, and ammonium. Limitation by the specific substrates elicited the up-regulation of their respective specific transporters. Wurch et al. (2011) examined changes in transcript abundances in *Aureococcus* under N- and P-stress and observed an up-regulation of alkaline phosphatase and 5′-nucleotidase under low-P conditions and up-regulation of an ammonium transporter, an acetamidase/formamidase, and two peptidases under N-limitation. Similarly, Agostoni and Erdner (2011) reported that an ammonium transporter was up-regulated in *A. lagunensis* under low ammonium conditions. These studies reveal the metabolic pathways that are induced by brown tides to acquire nutrients and confirm the importance of recycled inorganic or organic substrates in the nutrition of these species. These studies also provide a basis for the development biomarkers of N and P physiology in brown tide populations in the environment which may prove invaluable in assessing the role of limitation by specific nutrients (e.g., nitrogen) in regulating the formation and persistence of brown tide blooms.

6. Conclusions

Brown tides are perhaps the most notable examples of ecosystem disruptive algal blooms (EDABs) (Sunda et al., 2006; Buskey, 2008). The recurrent northeast U.S.A. brown tides caused by *Aureococcus anophagefferens* and the nearly 8-year Texas brown tide bloom caused by *Aureoumbra lagunensis* are also two of the most intensely studied EDABs. A combination of field observations, field experiments, laboratory culture experiments, as well as molecular and genomic studies have revealed the complex factors and mechanisms that promote these blooms.

The formation and persistence of brown tides appears to be linked to complex interactions among top down control by grazers, bottom up control by nutrients and light, grazer linked recycling of nutrients, and system flushing rates. Both species are poorly grazed by zooplankton and benthic herbivores, at least partly due to a polysacharride sheath surrounding the cells, and perhaps to cellular toxins, although none has yet been identified. The resultant decrease in grazing pressure appears to be an important factor promoting brown tide blooms. The role of nutrients is complex as both species have low maximum growth rates and compete poorly with other species, such as diatoms, at high concentrations of inorganic nutrients. Experimental evidence indicates that *Aureoumbra* cannot utilize nitrate while the growth of *Aureococcus* is disfavored relative to that of competing species by nitrate inputs. Consequently, blooms of either species require inputs of reduced nitrogen species supplied from nutrient recycling, either in the form of ammonium or a variety of organic N-containing compounds. New genomic data and a variety of laboratory culture and field experiments indicate that *Aureococcus* can utilize a wide range of organic N- and P-containing organic substrates and is also capable of growing heterotrophically on organic C.

New conceptual and numerical models and field observations indicate that brown tides are often preceded by blooms of high nutrient adapted phytoplankton (e.g. diatoms) whose growth is fueled by high levels of inorganic nutrients. This growth draws these nutrients down to low concentrations, and provides inputs of ammonium and organic nutrients from grazing-linked nutrient recycling. This facilitates a competitive shift in community dominance away from high-nutrient adapted species toward brown tide cells that accumulate in the water because of their low grazing mortality rates and ability to compete well at low nutrient availability. As brown tide blooms progress, positive feedbacks occur because poor grazing rates on the brown tide species decrease grazer-mediated recycling of nutrients, which further decreases available nutrients and favors competitive growth of the brown tide species. A decrease in the specific grazing rate on brown tide species with increasing nutrient limitation of growth rate should heighten this positive feedback, and further promote the formation of brown tide blooms. The blooms typically cause severe light attenuation, which further strengthens brown tide blooms because these algae grow maximally at low light levels and shading of benthic phototrophs increases the flux of regenerated nutrients from sediments. As with most algal blooms, the biomass and thus severity of brown tides is increased by inputs of nutrients from external sources, which are increasingly derived from anthropogenic sources.

Through the identification of factors that promote brown tides, this review suggests approaches that may be used in mitigating these recurrent blooms. These include: (1) a reduction in inputs of anthropogenic nutrients from municipal waste water, agricultural fertilizers, and the burning of fossil fuels; (2) the introduction and/or stocking of grazers (e.g. filter feeding mollusks) that are resistant to brown tides or can keep cell densities in check prior to bloom development; and (3) increasing the flushing rate of affected coastal lagoons via the widening of existing inlets and/or creation of new ones. Any one of these approaches would be a substantial undertaking and would require significant research and planning to successfully implement.

Acknowledgements

Writing this review was supported, in part, by award NA09NOS4780206 to C.J.G. from the National Oceanic and Atmospheric Administration Center for Sponsored Coastal Ocean Research's ECOHAB program and an award from New York Sea Grant to C.J.G. We are grateful for useful comments on this manuscript from Drs. Wayne Litaker, Jackie Collier, Sonya Dyhrman, and Sandra Shumway, and from Ms. Elyse Walker.[SS]

References

Agostoni, M., Erdner, D.L., 2011. Analysis of ammonium transporter and urease gene expression in *Aureoumbra lagunensis*. Harmful Algae 10, 549–556.

Anderson, D.M., Keafer, B.A., Kulis, D.M., Waters, R.M., Nuzzi, R., 1993. An immunofluorescent survey of the brown tide chrysophyte *Aureococcus anophagefferens* along the northeast coast of the United States. J. Plankton Res. 15, 563–580.

Bailey, J.C., Andersen, R.A., 1999. Analysis of clonal cultures of the brown tide algae *Aureococcus* and *Aureoumbra* (Pelagophyceae) using 18S rRNA, rbcL, and rubisco spacer sequences. J. Phycol. 35, 570–574.

Bersano, J.G.F., Buskey, E.J., Villareal, T.A., 2002. Viability of the Texas brown tide alga, Aureoumbra lagunensis, in fecal pellets of the copepod Acartia tonsa. Plankton Biol. Ecol. 49, 88–92.

Berg, G.M., Repeta, D.J., LaRoche, J., 2002. Dissolved organic nitrogen hydrolysis rates in axenic cultures of *Aureococcus anophagefferens* (Pelagophyceae): comparison with heterotrophic bacteria. Appl. Environ. Microbiol. 68, 401–404.

Berg, G.M., Shrager, J., Glöckner, G., Arrigo, K.R., Grossman, A.R., 2008. Understanding nitrogen limitation in *Aureococcus anophagefferens* (Pelagophyceae) through cDNA and qRT-PCR analysis. J. Phycol. 44, 1235–1249.

Bricelj, V.M., Kuenstner, S.H., 1989. Effects of the brown tide on the feeding physiology and growth of bay scallops and mussels. In: Cosper, E.M., Bricelj, V.M., Carpenter, E.J. (Eds.), Novel Phytoplankton Blooms: Causes and Impacts of Recurrent Brown Tides and Other Unusual Blooms. Springer, Berlin, pp. 491–509.

Bricelj, V.M., MacQuarrie, S.P., Schaffner, R.A., 2001. Differential effects of *Aureococcus anophagefferens* isolates (brown tide) in unialgal and mixed suspensions on bivalve feeding. Mar. Biol. 139, 605–615.

Bricelj, V.M., MacQuarrie, S.P., 2007. Effects of brown tide (*Aureococcus anophagefferens*) on hard clam *Mercenaria mercenaria* larvae and implications for benthic recruitment. Mar. Ecol. Prog. Ser. 331, 147–159.

Bricelj, V.M., MacQuarrie, S.P., Smolowitz, R., 2004. Concentration-dependent effects of toxic and non-toxic isolates of the brown tide alga *Aureococcus anophagefferens* on growth of juvenile bivalves. Mar. Ecol. Ser. Prog. 282, 101–114.

Buskey, E.J., 2008. How does eutrophication affect the role of grazers in harmful algal bloom dynamics? Harmful Algae 8, 152–157.

Buskey, E.J., Hyatt, C.J., 1995. Effects of the Texas (USA) brown tide alga on planktonic grazers. Mar. Ecol. Prog. Ser. 126, 285–292.

Buskey, E.J., Stockwell, D.A., 1993. Effects of a persistent brown tide on zooplankton populations in the Laguna Madre of south Texas. In: Smayda, T.J., Shimizu, Y. (Eds.), Toxic Phytoplankton Blooms in the Sea. Elsevier, Amsterdam, pp. 659–666.

Buskey, E.J., Wysor, B., Hyatt, C., 1998. The role of hypersalinity in the persistence of the Texas 'brown tide' in the Laguna Madre. J. Plankton Res. 20, 1553–1565.

Buskey, E.J., Montagna, P.A., Amos, A.F., Whitledge, T.E., 1997. Disruption of grazer populations as a contributing factor to the initiation of the Texas brown tide algal bloom. Limnol. Oceanogr. 42, 1215–1222.

Buskey, E.J., DeYoe, H., Jochem, F.J., Villareal, T.A., 2003. Effects of mesozooplankton removal and ammonium addition on planktonic trophic structure during a bloom of the Texas 'brown tide': a mesocosm study. J. Plankton Res. 25, 215–228.

Buskey, E.J., Liu, H.B., Collumb, C., Bersano, J.G.F., 2001. The decline and recovery of a persistent Texas brown tide algal bloom in the Laguna Madre (Texas, USA). Estuaries 24, 337–346.

Buskey, E.J., Stewart, S., Peterson, J., Collumb, C., 1996. Current Status and Historical Trends of Brown Tide and Red Tide Phytoplankton Blooms in the Corpus Christi Bay Nat Estuary Progr Study Area. Texas National Resource Conservation Commission, Austin, TX, 65 pp.

Caron, D.A., Gobler, C.J., Lonsdale, D.J., Buck, N.J., Cerrato, R.M., Schaffner, R.A., Rose, J.M., Taylor, G.T., Boissonneault, K.R., Mehran, R., 2004. Microbial herbivory on the brown tide alga, *Aureococcus anophagefferens*: results from natural ecosystems, mesocosms and laboratory experiments. Harmful Algae 3, 439–457.

Cerrato, R.M., Caron, D.A., Lonsdale, D.J., Rose, J.M., Schaffner, R.A., 2004. An experimental approach to examine the effect of the hard clam *Mercenaria mercenaria* on the development of blooms of the brown tide alga, *Aureococcus anophagefferens*. Mar. Ecol. Prog. Ser. 281, 93–108.

Cosper, E.M., Dennison, W.C., Carpenter, E.J., Bricelj, V.M., Mitchell, J.G., Kuenstner, S.H., Colflesh, D.C., Dewey, M., 1987. Recurrent and persistent brown tide blooms perturb coastal marine ecosystem. Estuaries 10, 284–290.

Cosper, E.M., Dennison, W., Milligan, A., 1989. An examination of environmental factors important to initiating and sustaining brown tide blooms. In: Cosper, E.M., Bricelj, V.M., Carpenter, E.J. (Eds.), Novel Phytoplankton Blooms: Causes and Impacts of Recurrent Brown Tides and Other Unusual Blooms. Springer, Berlin, pp. 317–340.

Deonarine, S.N., Gobler, C.J., Lonsdale, D.J., Caron, D.A., 2006. The role of zooplankton in the occurrence of harmful brown tide blooms (*Aureococcus anophagefferens*) in US mid-Atlantic estuaries. Aquat. Microb. Ecol. 44, 181–195.

DeYoe, H.R., Suttle, C.A., 1994. The inability of the Texas brown tide alga to use nitrate and the role of nitrogen in the initiation of a persistent bloom of this organism. J. Phycol. 30, 800–806.

DeYoe, H.R., Buskey, E.J., Jochem, F.J., 2007. Physiological responses of *Aureococcus lagunensis* and *Synechococcus* sp. to nitrogen addition in a mesocosm study. Harmful Algae 6, 48–55.

DeYoe, H.R., Chan, A.M., Suttle, C.A., 1995. Phylogeny of *Aureococcus anophagefferens* and a morphologically similar bloom-forming alga from Texas as determined by 18S ribosomal RNA sequence analysis. J. Phycol. 31, 413–418.

DeYoe, H.R., Stockwell, D.A., Bidigare, R.R., Latasa, M., Johnson, P.W., Hargraves, P.E., Suttle, C.A., 1997. Description and characterization of the algal species *Aureoumbra lagunensis* gen. et sp. nov. and referral of *Aureoumbra* and *Aureococcus* to the Pelagophyceae. J. Phycol. 33, 1042–1048.

Dzurica, S., Lee, C., Cosper, E.M., Carpenter, E.J., 1989. Role of environmental variables, specifically organic compounds and nutrients, in the growth of the chrysophyte Aureococcus anophagefferens. In: Cosper, E.M., Bricelj, V.M., Carpenter, E.J. (Eds.), Novel Phytoplankton Blooms: Causes and Impacts of Recurrent Brown Tides and Other Unusual Blooms. Springer, New York, pp. 229–252.

Frazier, A.N.D., Rowe, J.I.M., Rentz, C., Gobler, C.J., Wilhelm, S.W., 2007. Bacterial lysis of *Aureococcus anophagefferens* CCMP 1784. J. Phycol. 43, 461–465.

Gainey, L.F., Shumway, S.E., 1991. The physiological effect of *Aureococcus anophagefferens* (brown tide) on the lateral cilia of bivalve mollusks. Biol. Bull. 181, 298–306.

Gallager, S.M., Stoecker, D.K., Bricelj, V.M., 1989. Effects of the brown tide algae on growth, feeding physiology and locomotory behavior of scallop larvae (Argopecten irradians). In: Cosper, E.M., Bricelj, V.M., Carpenter, E.J. (Eds.), Novel Phytoplankton Blooms: Causes and Impacts of Recurrent Brown Tides and Other Unusual Blooms. Springer, Berlin, pp. 511–542.

Gastrich, M.D., Anderson, O.R., Cosper, E.M., 2002. Viral-like particles (VLPS) in the alga, *Aureococcus anophagefferens* (Pelagophyceae), during 1999–2000 brown tide blooms in Little Egg Harbor, New Jersey. Estuaries 25, 938–943.

Gastrich, M.D., Bell, J.L., Gobler, C.J., Anderson, O.R., Wilhelm, S.W., 2004. Viruses as potential regulators of regional brown tide blooms caused by the alga, *Aureococcus anophagefferens:* a comparison of bloom years 1999–2000 and 2002. Estuaries 27, 112–119.

Giner, J.L., Li, X.Y., Boyer, G.L., 2001. Sterol composition of *Aureoumbra lagunensis*, the Texas brown tide alga. Phytochemistry 57, 787–789.

Glibert, P.M., Wazniak, C.E., Hall, M.R., Sturgis, B., 2007. Seasonal and interannual trends in nitrogen and brown tide in Maryland's coastal bays. Ecol. Appl. 17, S79–S87.

Gobler, C.J., Sañudo-Wilhelmy, S.A., 2001a. Effects of organic carbon, organic nitrogen, inorganic nutrients, and iron additions on the growth of phytoplankton and bacteria during a brown tide bloom. Mar. Ecol. Prog. Ser. 209, 19–34.

Gobler, C.J., Sañudo-Wilhelmy, S.A., 2001b. Temporal variability of groundwater seepage and brown tide blooms in a Long Island embayment. Mar. Ecol. Prog. Ser. 217, 299–309.

Gobler, C.J., Renaghan, M.J., Buck, N.J., 2002. Impacts of nutrients and grazing mortality on the abundance of *Aureococcus anophagefferens* during a New York brown tide bloom. Limnol. Oceanogr. 47, 129–141.

Gobler, C.J., Deonarine, S., Leigh-Bell, J., Gastrich, M.D., Anderson, O.R., Wilhelm, S.W., 2004a. Ecology of phytoplankton communities dominated by *Aureococcus anophagefferens*: the role of viruses, nutrients, and microzooplankton grazing. Harmful Algae 3, 471–483.

Gobler, C.J., Boneillo, G.E., Debenham, C., Caron, D.A., 2004b. Nutrient limitation, organic matter cycling, and plankton dynamics during an *Aureococcus anophagefferens* bloom in Great South Bay, NY. Aquat. Microb. Ecol. 35, 31–43.

Gobler, C.J., Downes Gastrich, M., Anderson, O.R., Wilhelm, S.W., 2007. Ecological aspects of viral infection and lysis in the harmful brown tide alga, *Aureococcus anophagefferens*. Aquat. Microb. Ecol. 47, 25–36.

Gobler, C.J., 32 co-authors, 2011. Niche of harmful alga Aureococcus anophagefferens revealed through ecogenomics. Proc. Natl. Acad. Sci. U.S.A. 108, 4352–4357.

Goericke, R., 1998. Response of phytoplankton community structure and taxon-specific growth rates to seasonally varying physical forcing in the Sargasso Sea off Bermuda. Limnol. Oceanogr. 43, 921–935.

Greenfield, D.I., Lonsdale, D.J., 2002. Mortality and growth of juvenile hard clams *Mercenaria mercenaria* during brown tide. Mar. Biol. 141, 1045–1050.

Hairston, N.G., Holtmeier, C.L., Lampert, W., Weider, L.J., Post, D.M., Fisher, J.M., Caceres, C.E., Gaedke, U., 2002. Natural selection for grazer resistance to toxic cyanobacteria: evolution of phenotypic plasticity? Evolution 55, 2203–2214.

Hardy, C., 1976. A preliminary description of the Peconic Bay Estuary. Marine Science Research. Center, SUNY, Stony Brook, New York. Special Report No. 76-4.

Harke, M.J., Gobler, C.J., Shumway, S.E., 2011. Suspension feeding by the Atlantic slipper limpet (Crepidula fornicata) and the northern quahog (Mercenaria mercenaria) in the presence of cultured and wild populations of the harmful brown tide alga, Aureococcus anophagefferens. Harmful Algae 10, 503–511.

Hattenrath-Lehmann, T.K., Gobler, C.J., 2011. Allelopathic inhibition of competing phytoplankton by North American strains of the toxic dinoflagellate, Alexandrium fundyense: evidence from field experiments, laboratory experiments, and bloom events. Harmful Algae 11, 106–116.

Heisler, J.P., Glibert, P.M., Burkholder, J.A., Anderson, D.M., Cochlan, W., Dennison, W., Dortch, Q., Gobler, C.J., Heil, C., Humphries, E., Lewitus, A., Magnien, R., Marshall, H., Sellner, S., Stockwell, D., Stoecker, D., Suddleson, M., 2008. Eutrophication and harmful algal blooms: a scientific consensus. Harmful Algae 8, 3–13.

Keller, A.A., Rice, R.L., 1989. Effects of nutrient enrichment on natural populations of the brown tide phytoplankton *Aureococcus anophagefferens* (Chrysophyceae). J. Phycol. 25, 636–646.

Kraeuter, J.N., Klinck, J.M., Powell, E.N., Hofmann, E.E., Buckner, S.C., Grizzle, R.E., Bricelj, V.M., 2008. Effects of the fishery on the northern quahog (=hard clam, Mercenaria mercenaria L.) population in Great South Bay, New York: A modeling study. J. Shellfish Res. 27, 653–666.

LaRoche, J., Nuzzi, R., Waters, R., Wyman, K., Falkowski, P.G., Wallace, D.W.R., 1997. Brown tide blooms in Long Island's coastal waters linked to interannual variability in groundwater flow. Glob. Change Biol. 3, 397–410.

Liu, H., Buskey, E.J., 2000. The exopolymer secretions (EPS) layer surrounding *Aureoumbra lagunensis* cells affects growth, grazing, and behavior of protozoa. Limnol. Oceanogr. 45, 1187–1191.
Liu, H., Laws, E.A., Villareal, T.A., Buskey, E.J., 2001. Nutrient-limited growth of *Aureoumbra lagunensis* (Pelagophyceae), with implications for its capability to outgrow other phytoplankton species in phosphate-limited environments. J. Phycol. 37, 500–508.
Lomas, M.W., Glibert, P.M., Berg, G.M., 1996. Characterization of nitrogen uptake by natural populations of *Aureococcus anophagefferens (*Chrysophyceae) as a function of incubation duration, substrate concentration, light, and temperature. J. Phycol. 32, 907–916.
Lomas, M.W., Kana, T.M., MacIntyre, H.L., Cornwell, J.C., Nuzzi, R., Waters, R., 2004. Interannual variability of *Aureococcus anophagefferens* in Quantuck Bay, Long Island: natural test of the DON hypothesis. Harmful Algae 3, 389–402.
Lonsdale, D.J., Cosper, E.M., Kim, W.-S., Doall, M.H., Divadeenam, A., Jonasdottir, S.H., 1996. Food web interactions in the plankton of Long Island bays, with preliminary observations on brown tide effects. Mar. Ecol. Prog. Ser. 134, 247–263.
MacIntyre, H.L., Lomas, M.W., Cornwell, J., Suggett, D.J., Gobler, C.J., Koch, E.W., Kana, T.M., 2004. Mediation of benthic-pelagic coupling by microphytobenthos: an energy- and material-based model for initiation of blooms of *Aureococcus anophagefferens*. Harmful Algae 3, 403–437.
Milligan, K.L.D., Cosper, E.M., 1994. Isolation of virus capable of lysing the brown tide microalga, *Aureococcus anophagefferens*. Science 266, 805–807.
Milligan, A.J., Cosper, E.M., 1997. Growth and photosynthesis of the 'brown tide' microalga *Aureococcus anophagefferens* in subsaturating constant and fluctuating irradiance. Mar. Ecol. Prog. Ser. 153, 67–75.
Mitra, A., Flynn, K., 2006. Promotion of harmful algal blooms by zooplankton predatory activity. Biol. Lett. 2, 194–197.
Montagna, P.A., Stockwell, D.A., Kalke, R.D., 1993. Dwarf surf clam *Mulinea lateralis* (Say, 1822) populations and feeding during the Texas brown tide. J. Shellfish Res. 12, 833–842.
Muhlstein, H.I., Villareal, T.A., 2007. Organic and inorganic nutrient effects on growth rate-irradiance relationships in the Texas brown-tide alga *Aureoumbra lagunensis* (Pelagophyceae). J. Phycol. 43, 1223–1226.
Mulholland, M.R., Gobler, C.J., Lee, C., 2002. Peptide hydrolysis, amino acid oxidation, and nitrogen uptake in communities seasonally dominated by *Aureococcus anophagefferens*. Limnol. Oceanogr. 47, 1094–1108.
Mulholland, M.R., Boneillo, G.E., Bernhardt, P.W., Minor, E.C., 2009. Comparison of nutrient and microbial dynamics over a seasonal cycle in a mid-atlantic coastal lagoon prone to Aureococcus anophagefferens (Brown Tide) Blooms. Estuaries Coasts 32, 1176–1194.
Newell, R.I.E., Tettelbach, S.T., Gobler, C.J., Kimmel, D.G., 2009. Relationships between reproduction in suspension-feeding hard clams, *Mercenaria mercenaria*, and phytoplankton community structure. Mar. Ecol. Prog. Ser. 387, 179–196.
Nuzzi, R., Waters, R.A., 2004. Long-term perspective on the dynamics of brown tide blooms in Long Island coastal bays. Harmful Algae 3, 279–293.
Olsen, P.S., 1989. Development and distribution of a brown-water algal bloom in Barnegat Bay, New Jersey with perspective on resources and other red tides in the region. In: Cosper, E.M., Bricelj, V.M., Carpenter, E.J. (Eds.), Novel Phytoplankton Blooms: Causes and Impacts of Recurrent Brown Tides and Other Unusual Blooms. Springer, Berlin, pp. 189–212.
Ong, H.C., Wilhelm, S.W., Gobler, C.J., Bullerjahn, G., Jacobs, M.A., Sims, E.H., Gillett, W.G., Zhou, Y., Haugen, E., Rocap, G., Cattolico, R.A., 2010. Analyses of the complete chloroplast genome sequences of two members of the Pelagophyceae: *Aureococcus anophagefferens* CCMP1984 and *Aureoumbra lagunensis* CCMP1507. J. Phycol. 46, 602–615.
Onuf, C.P., 1996. Seagrass responses to long-term light reduction by brown tide in upper Laguna Madre, Texas: distribution and biomass patterns. Mar. Ecol. Prog. Ser. 138, 219–231.
Padilla, D.K., Doall, M.H., Gobler, C.J., Hartson, A., O'Boyle, K., 2006. Brown tide alga, *Aureococcus anophagefferens*, can affect growth but not survivorship of *Mercenaria mercenaria* larvae. Harmful Algae 5, 736–748.
Popels, L.C., Coyne, K.J., Forbes, R., Pustizzi, F., Gobler, C.J., Cary, S.C., Hutchins, D.A., 2003. The use of quantitative polymerase chain reaction for the detection and enumeration of the harmful alga *Aureococcus anophagefferens* in environmental samples along the United States east coast. Limnol. Oceanogr. Methods 1, 92–102.
Popels, L.C., MacIntyre, H.L., Warner, M.E., Zhang, Y.H., Hutchins, D.A., 2007. Physiological responses during dark survival and recovery in Aureococcus anophagefferens (Pelagophyceae). J. Phycol. 43, 32–42.
Porter, K., 1976. Enhancement of algal growth and productivity by grazing zooplankton. Science 192, 1332–1333.
Probyn, T., Pitcher, G., Pienaar, R., Nuzzi, R., 2001. Brown tides and mariculture in Saldanha Bay, South Africa. Mar. Poll. Bull. 42, 405–408.
Probyn, T.A., Bernard, S., Pitcher, G.C., Pienaar, R.N., 2010. Ecophysiological studies on *Aureococcus anophagefferens* blooms in Saldanha Bay, South Africa. Harmful Algae 9, 123–133.
Raven, J.A., Kubler, J.E., 2002. New light on the scaling of metabolic rate with the size of algae. J. Phycol. 38, 11–16.
Robbins, H.M., Bricelj, V.M., Ward, J.E., 2010. In vivo effects of brown tide on the feeding function of the gill of the Northern Quahog *Mercenaria mercenaria* (Bivalvia: Veneridae). Biol. Bull. 219, 61–71.
Rowe, J.M., Dunlap, J.R., Gobler, C.J., Downes Gastrich, M., Anderson, O.R., Wilhelm, S.W., 2008. Isolation of a non-phage-like lytic virus infecting *Aureococcus anophagefferens*. J. Phycol. 44, 71–76.
Sherr, E.B., Sherr, B.F., 2002. Significance of predation by protists in aquatic microbial food webs. Antonie van Leeuwenhoek 81, 293–308.
Sieburth, J.M., Johnson, P.W., Hargraves, P.E., 1988. Ultrastructure and ecology of *Aureococcus anophagefferens* gen. et sp. nov. (Chrysophyceae): the dominant picoplankter during a bloom in Narragansett Bay, Rhode Island, summer 1985. J. Phycol. 24, 416–425.
Sieracki, M.E., Gobler, C.J., Cucci, T., Their, E., Hobson, I., 2004. Pico- and nanoplankton dynamics during bloom initiation of *Aureococcus* in a Long Island, NY bay. Harmful Algae 3, 459–470.
Smayda, T.J., 2008. Complexity in the eutrophication-harmful algal bloom relationship, with comment on the importance of grazing. Harmful Algae 8, 140–151.
Smith, J.K., Lonsdale, D.J., Gobler, C.J., Caron, D.A., 2008. Feeding behavior and development of *Acartia tonsa* nauplii on the brown tide alga *Aureococcus anophagefferens*. J. Plankton Res. 30, 937–950.
Sunda, W.G., Graneli, E., Gobler, C.J., 2006. Positive feedback and the development and persistence of ecosystem disruptive algal blooms. J. Phycol. 42, 963–974.
Sunda, W.G., Hardison, D.R., 2007. Ammonium uptake and growth limitation in marine phytoplankton. Limnol. Oceanogr. 52, 2496–2506.
Sunda, W.G., Hardison, D.R., 2010. Evolutionary tradeoffs among nutrient acquisition, cell size, and grazing defense in marine phytoplankton promote ecosystem stability. Mar. Ecol. Prog. Ser. 401, 63–76.
Sunda, W.G., Shertzer, K. Modeling ecosystem disruptive algal blooms: Positive feedback mechanisms. Mar. Ecol. Prog. Ser., in press.
Tang, Y.Z., Gobler, C.J., 2010. Allelopathic effects of Cochlodinium polykrikoides isolates and blooms from the estuaries of Long Island, New York, USA on co-occurring phytoplankton. Mar. Ecol. Prog. Ser. 406, 19–31.
Tang, Y.Z., Gobler, C.J., 2011. Allelopathic effects of the green macroalgae, *Ulva lactuca*, on seven common harmful algal bloom species. Harmful Algae 10, 480–488.
Taylor, G.T., Gobler, C.J., Sañudo-Wilhelmy, S.A., 2006. Nitrogen speciation and concentrations of nitrogen as determinants of Brown Tide (*Aureococcus anophagefferens*, pelagophyceae) initiation: An experimental evaluation. Mar. Ecol. Prog. Ser. 312, 67–83.
Trice, T.M., Glibert, P.M., Lea, C., Van Heukelem, L., 2004. HPLC pigment records provide evidence of past blooms of *Aureococcus anophagefferens* in the Coastal Bays of Maryland and Virginia, USA. Harmful Algae 3, 295–304.
Villareal, T.A., Chirichella, T., Buskey, E.J., 2004. Regional distribution of the Texas Brown Tide (Aureoumbra lagunensis) in the Gulf of Mexico. In: Steidinger, K.A., Landsberg, J.H., Tomas, C.R., Vargo, G.A. (Eds.), Harmful Algae. 2002. Florida Fish and Wildlife Conser. Comm., FL Inst. Oceanogr., IOC-UNESCO, St. Petersburg, pp. 374–376.
Ward, L.A., Montagna, P.A., Kalke, R.D., Buskey, E.J., 2000. Sublethal effects of Texas brown tide on *Streblospio benedicti* (Polychaeta) larvae. J. Exp. Mar. Biol. Ecol. 248, 121–129.
Wilson, R., 1995. Aspects of tidal and subtidal flushing within the Peconic Bays Estuary. In: McElroy, A. (Ed.), Proc. Brown Tide Summit, NYSGI-W-95-001. New York Sea Grant Inst., NY, pp. 53–56.
Wurch, L., Haley, S., Orchard, E., Gobler, C.J., Dhyrman, S.D., 2011. Nutrient-regulated transcriptional responses in the brown tide forming alga *Aureococcus anophagefferens*. Environ. Microbiol. 13, 468–481.
Yentsch, C.S., Phinney, D.A., Shapiro, L.P., 1989. Absorption and fluorescent characteristics of the brown tide chrysophyte: its role on light reduction in coastal marine environments. In: Cosper, E.M., Bricelj, V.M., Carpenter, E.J. (Eds.), Novel Phytoplankton Blooms: Causes and Impacts of Recurrent Brown Tides and other Unusual Blooms. Springer, Berlin, pp. 77–83.